A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived analysis

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Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ-VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A runtime operator audit of the seed-0 20.1M ViT checkpoint finds no incompatible learned operation, and local reconstructions reproduce its deployed rational and bilinear branches near the float32 precision floor. Across three independently trained 20.1M ViT policies, closed-loop LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical trials per checkpoint. A preregistered elementwise prediction-mean ensemble reaches $467/500 = 93.4\%$ at the cost of three forward passes. Separately, three jointly trained checkpoints spanning 40 familiar LIBERO tasks reach $70.35\% \pm 1.32\%$ task-macro success, compared with $69.12\% \pm 2.33\%$ for parameter-matched conventional controls under the same recipe. Their frozen three-pass mean-prediction ensemble reaches 74.75% with every suite at or above 50%. At equal aggregate updates, a frozen seed-0 joint checkpoint retains 95.16% of four suite specialists' pooled success. On the three LIBERO-Object checkpoints, a prospectively frozen paired test on 100 unique familiar-task states finds $247/300 = 82.3\%$ correct-instruction success, versus $72/300 = 24.0\%$ with a co-present control instruction and $78/300 = 26.0\%$ with an empty instruction. On disjoint episodes, zeroing only the learned post-lookup token vectors drops success from $263/300 = 87.7\%$ to $63/300 = 21.0\%$, with positive gaps for all ten task aggregates. On a separate three-checkpoint convolutional instantiation, independent reconstruction covers all $4{,}608$ joint-attention head-input cases, all 384 joint-FFN module-input cases, and all 384 complete joint-block input cases. The respective worst relative errors are 6.36×10^{-16}, 1.39×10^{-15}, and 1.5024×10^{-7}. Exact-attention records naming the same three ViT checkpoint artifacts derive the softmax-free attention decomposition and numerically replay every attention module on one fixed corrected cached input per checkpoint, with worst relative error 2.016119×10^{-7}. A separate sequential replay independently rebuilds the learned path through four vision and eight joint blocks to the linear action output on 48 fixed inputs, with worst relative error 3.9822×10^{-7} against a frozen 10^{-3} gate. In a seed-0 ViT checkpoint, all 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. Separately, prospectively fixed rank-96 visual

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projectors retain 95.2–98.2% of baseline successes on tasks absent from corrected projector construction, while three equal-rank random projectors retain only 3.3–13.6%. The evidence establishes strong single-suite and four-suite familiar-task capability and exact layerwise access to trained attention coefficients, together with bounded familiar-task instruction necessity. It does not establish unseen-task generalization, counterfactual goal following, or an input-general compact symbolic decomposition of the full policy.

1 Introduction

Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains millions of learned numbers, but it does not directly reveal which combinations of image regions, instruction tokens, and robot state produce an action. Analysts therefore usually collect activations on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the same structure is present in the weights and whether it causes behavior.

Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. Decomposing that table can rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. This is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

> *Can a compact VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?*

We test capability and structural analysis in two fully rational encoder instantiations. The same three ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate convolutional instantiation tests whether the structural certificates transfer across visual encoders.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate three fully rational 20.1M ViT policies on $1{,}500$ canonical LIBERO-Object trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches $467/500 = 93.4\%$. Three jointly trained checkpoints average $70.35\% \pm 1.32\%$ over 40 familiar tasks, 1.23 points above matched conventional controls. Their fixed ensemble reaches 74.75%. A frozen seed-0 comparison retains 95.16% of four suite specialists.
3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor. In a separate

closed-loop audit, learned rank-96 visual projectors retain 95.2–98.2% of baseline successes, versus 3.3–13.6% for random projectors.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j}\right)\bar{x}_i \bar{x}_j. \tag{1}$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top)/s\right] V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. Let $m(x) = d^{-1}\sum_{j=1}^{d} x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \qquad \mathrm{RNorm}(x) = \frac{x}{\sqrt{s_0}}\, r(v). \tag{3}$$

Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Linear, bilinear, residual, and normalization operations update this pair. The model therefore keeps input-specific rescaling while remaining exactly representable as a rational tensor network. Training and deployment use r itself. The exactness claims below therefore concern faithful evaluation of the deployed rational computation, not uniform agreement with RMSNorm. Appendix A.5 certifies pole-free denominators and deployed arithmetic fidelity while retaining that boundary.

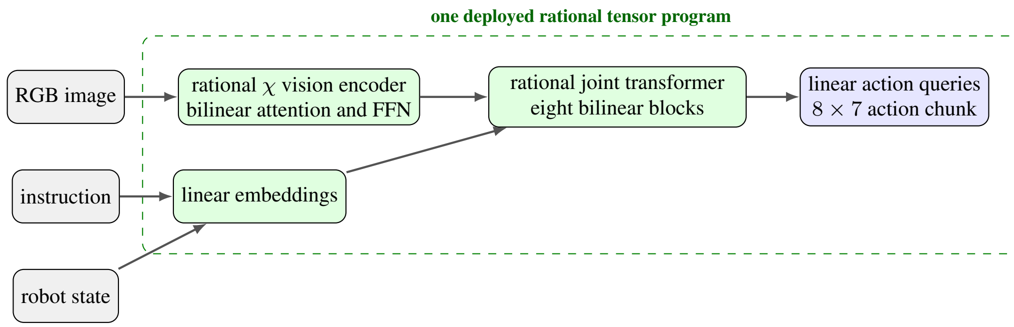

Figure 1: **Deployed χ-VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

2.2 Architecture and training

Both encoder variants follow the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional files receive the joint-attention and rational-normalizer certificates reported in Appendix A.5.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M ViT checkpoint at runtime and reconstruct representative operations from saved weights.

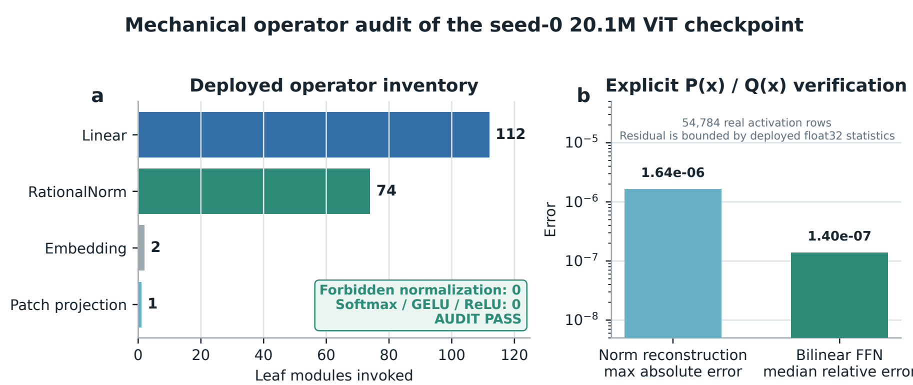

Figure 2: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 50 canonical trials per task
- a 280-step cap
- three independently trained ViT checkpoints
- 500 rollouts per checkpoint and 1,500 rollouts in total

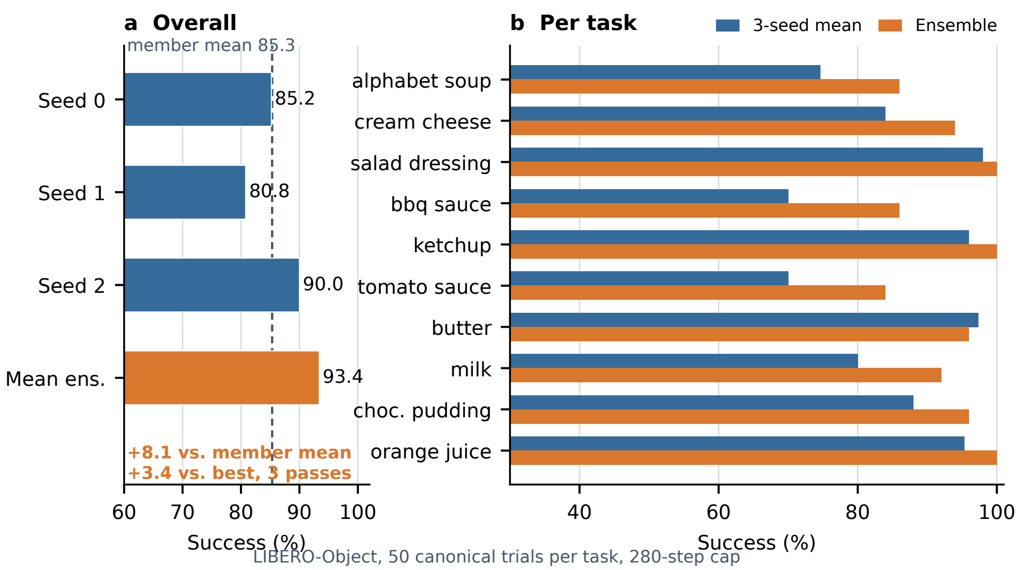

Figure 3: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ-VLA, rational ViT	20.1M	1 pass	$\mathbf{85.3\% \pm 4.6\%}$
χ-VLA, prediction-mean ensemble	3×20.1M	3 passes	$\mathbf{93.4\%}$

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Elementwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points above the member mean and 3.4 points above the strongest member. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task index, and protocol field is retained. The artifact manifest binds each immutable result file and recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic gaps are descriptive rather than a matched-hardware ensemble ablation.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

3.4 One policy across four familiar suites

We next train one 20.1M χ-VLA checkpoint jointly on the ten tasks from each of LIBERO-Object, LIBERO-Spatial, LIBERO-Goal, and LIBERO-10. The balanced recipe uses $160{,}000$ updates, equal to the aggregate $4 \times 40{,}000$ updates used by four separately trained suite specialists. Evaluation uses 50 canonical trials per task, so each checkpoint receives $2{,}000$ closed-loop trials.

Policy	Object	Spatial	Goal	LIBERO-10	40-task macro
Joint χ-VLA, three-seed mean	80.9%	73.1%	81.9%	45.5%	$\mathbf{70.35\% \pm 1.32\%}$
Matched conventional, three-seed mean	79.2%	77.0%	84.4%	35.9%	$69.12\% \pm 2.33\%$
Fixed three-checkpoint mean ensemble	88.6%	74.0%	85.6%	50.8%	$\mathbf{74.75\%}$
Joint χ-VLA, seed 0	83.6%	73.0%	82.0%	44.8%	70.85%
Four suite specialists, seed 0	83.0%	80.0%	88.0%	46.8%	74.45%

Table 2: **Closed-loop breadth over 40 familiar tasks.** The first row is the mean across three jointly trained checkpoints, with sample standard deviation across their 40-task macro scores. The matched conventional row uses the same balanced recipe. The ensemble averages normalized χ-VLA action predictions and costs three forward passes. The final two rows form the separately frozen seed-0 retention comparison.

The three joint checkpoints score 70.85%, 71.35%, and 68.85%, and their seed-mean success is at least 50% on $32/40$ tasks. In the separately frozen same-recipe comparison, three parameter-matched conventional checkpoints average $69.12\% \pm 2.33\%$. The χ-VLA mean is 1.23 percentage points higher and passes the fixed requirement of being within five points of the conventional mean. The model sizes differ by only $1{,}280$ parameters, or 0.006%. In the separately frozen mean-prediction ensemble, success rises from the 70.35% member mean to 74.75%, with every suite at or above 50% and $34/40$ tasks at or above 50%. Its protocol and gates were fixed after the member checkpoints were selected but before ensemble evaluation. In the separately frozen seed-0 comparison, the single joint checkpoint retains $70.85/74.45 = 95.16\%$ of the four specialists' pooled task-macro success, above the fixed 85% retention threshold. The suite ratios are 100.7%, 91.3%, 93.2%, and 95.7% for Object, Spatial, Goal, and LIBERO-10. This certificate compares familiar tasks seen during training. It establishes multi-suite capacity and parameter sharing, not transfer to an unseen task, instruction, embodiment, or domain.

3.5 Prospectively frozen closed-loop instruction necessity

High task success alone does not show that a policy uses its instruction. Before these outcomes were observed, we fixed a paired test on canonical episodes 40–49 for every task and the same three checkpoints. Each checkpoint-state pair was rolled out for the full 280-step horizon under the correct familiar instruction, one deterministic outcome-independent co-present control instruction, and an empty instruction. All conditions shared the exact initial physical input and their order was chosen by a frozen hash. Co-presence was verified from simulator observation fields and does not guarantee unoccluded camera visibility. The empty string encodes tokenizer BOS plus 31 unmasked PAD tokens, together with the model's learned common BOS.

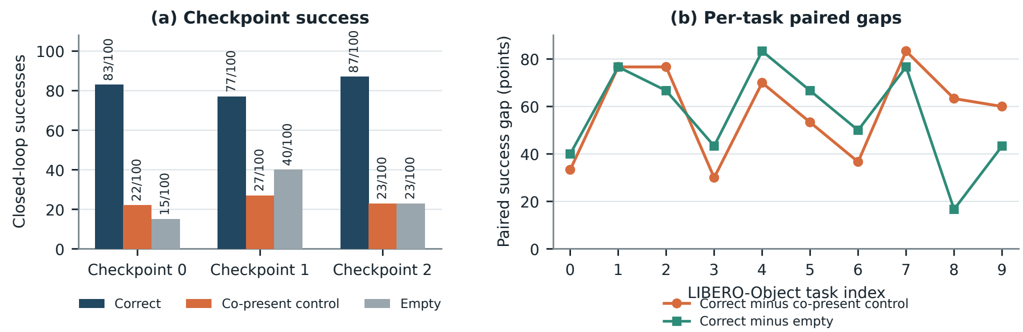

Figure 4: **Closed-loop instruction necessity on familiar tasks.** Left: exact successes among 100 paired canonical states per checkpoint. Right: correct-instruction success minus each control, pooling the three checkpoint outcomes for each task. All ten task gaps are positive for both controls at the task-aggregate level.

Across 100 unique canonical states and three checkpoints, the correct instruction succeeds on $247/300 = 82.3\%$ of checkpoint-state pairs. The co-present control succeeds on $72/300 = 24.0\%$ and the empty instruction on $78/300 = 26.0\%$, giving paired gaps of 58.3 and 56.3 points. Correct-instruction success is $83/100$, $77/100$, and $87/100$ by checkpoint, and both paired gaps are positive for every checkpoint aggregate. All ten pooled task aggregates are strictly positive for both controls. One-sided exact McNemar tests give $p = 2.02 \times 10^{-45}$ and $p = 1.52 \times 10^{-44}$. These pooled tests use 300 checkpoint-state pairs and are not cluster-robust inference. A frozen 20,000-draw task-stratified state-cluster bootstrap resamples 100 unique task-episode states while retaining all three checkpoint outcomes. Its 95% intervals are $[51.0, 65.7]$ and $[50.0, 62.3]$ points. Every prospectively frozen gate passes.

This establishes instruction necessity for familiar LIBERO-Object prompts, objects, scenes, tasks, and checkpoints. Appendix A.1 gives a disjoint paired test that isolates the learned lexical-embedding path. Neither test establishes counterfactual goal following, paraphrase robustness, unseen-object or compositional grounding, cross-suite or physical transfer, or a benefit unique to tensor decomposability.

Within its stated boundary, this closes the possibility that the policy's familiar-task success is independent of its instruction. It does not turn the specialist benchmark into a general language grounding result.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. On one fixed corrected cached input per checkpoint, we numerically replay all four vision and eight joint attention modules in each of three records that name the same ViT checkpoint artifacts as the capability results. This covers 36 module-input cases and 384 architectural heads. The worst relative ℓ_2 error is 2.016119×10^{-7} and the worst absolute difference is 2.44×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients. The algebraic identity is input-general. The one-input-per-checkpoint records provide a layerwise numerical replay audit, not an exact end-to-end policy decomposition.

We separately certify three convolutional checkpoints from the same rational architecture family. Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs per checkpoint yield $4{,}608$ head-input cases. Every output is finite and the worst relative ℓ_2 error is 6.36×10^{-16}, below the frozen 10^{-6} gate. On the same inputs, independent CP-factor evaluation covers all 384 joint-FFN module-input cases at worst relative error 1.3871×10^{-15} against a 10^{-10} gate. Independent complete-block equations cover all 384 block-input cases at worst relative error 1.5024×10^{-7} against a 10^{-4} gate. The complete-block comparison crosses a deployed PyTorch float32 block call and a NumPy float64 reconstruction. It is input-conditioned and per-block, not a sequential end-to-end reconstruction of the joint transformer or policy.

A separately frozen ViT audit partitions joint attention into vision, instruction, robot-state, and action-query source tokens. Across 128 deterministic, official-task-balanced inputs for each of the three capability checkpoints, all $36{,}864$ head-input cases and $147{,}456$ source-group evaluations are finite, with worst reconstruction error 5.0741×10^{-7} against a 10^{-6} gate. Across the $3{,}072$ projected module-input cases, mean coherent-energy shares are 67.7% vision, 19.2% robot state, 7.9% action query, and 5.1% instruction. These input-conditioned magnitudes are descriptive rather than causal importance. Familiar-prompt permutations are also descriptive and do not establish grounding or closed-loop success.

Finally, a prospectively frozen sequential replay reaches the learned action output on the same three capable ViT checkpoints. For 16 deterministic, official-task-balanced corrected inputs per checkpoint, an independent NumPy float64 evaluation starts from the exact prepared model tensors and rebuilds the learned patch projection, four vision blocks, multimodal embeddings and projections, eight joint blocks, final RationalNorm, and linear action head. Across 48 inputs and 816 recorded stage evaluations, the worst final-action relative ℓ_2 error is 3.9822×10^{-7} against a frozen 10^{-3} gate. Every manual PyTorch traversal is bitwise identical to an unmodified full model call. This is an input-conditioned full learned-forward numerical certificate, not one compact symbolic contraction or an input-general proof. It excludes preprocessing, fixed action unnormalization, chunk execution, simulator dynamics, and success.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14}.

4.2 Trained policy result

The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer's computation.

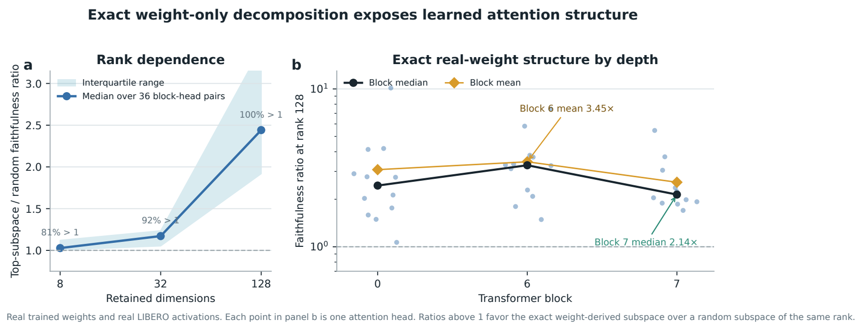

Figure 5: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent model-level replicates.

This extends exact weight-only construction through individual attention layers. It is not an exact compact symbolic end-to-end decomposition of the complete policy. Symbolic composition across all eight blocks still faces rapid degree and representation growth.

A separate intervention tests behavioral sufficiency at the visual-to-joint interface rather than deriving a projector from the exact attention factors. An action-Jacobian projector estimated on official Object tasks 0–3, with rank 96 fixed before corrected outcomes, retains $259/272$ full-policy successes on tasks 8–9 across three ViT checkpoints. These tasks are absent from corrected projector construction. An activation-energy projector retains $267/272$, while three equal-rank random projectors retain $37/272$, $9/272$, and $22/272$. All ten tasks appeared during policy training. This establishes a structured low-dimensional visual bottleneck relative to random projectors, not unseen-task generalization, unique action-Jacobian specificity, or a causal link to the exact weight factors. Appendix A.4 reports the paired controls and stability intervals.

5 Related work

Capability, mechanisms, and robustness. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2], while χ-VLA studies a compact algebraically restricted policy on familiar single-suite and jointly trained four-suite tasks. Polynomial and tensor networks provide the multilinear foundation [12–14]. Bilinear MLPs and χ-nets expose low-rank features and compositional circuits from trained weights [3, 4]. We extend that setting to closed-loop control and exact reduced softmax-free attention. Bilinear policy analysis and recent activation, circuit, and rollout studies make our distinction empirical rather than universal [5–8]. Because LIBERO success can coexist with weak robustness and language use [9–11], our claims remain bounded to familiar tasks from the training suites.

6 Limitations and conclusion

χ-VLA shows that strong single-suite control and substantial four-suite familiar-task breadth can coexist with exact access to trained structure. The paired controls establish familiar-task instruction necessity, not counterfactual goal following, unseen-task generalization, or physical transfer. The learned path is replayed on fixed inputs, not given as one compact symbolic contraction or input-general proof, and the strongest ensemble costs three passes. Common ViT checkpoints connect capability, exact attention, and reduced-Gram analysis, while separate convolutional files replicate the attention equation certificate. We do not claim a coefficient-edit interface. The anonymous artifact and Appendix A.2 provide full reproducibility details.

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A Supporting results and decisions

A.1 Causal isolation of the learned instruction-embedding path

Token IDs are lookup addresses. A learned embedding table turns those addresses into vectors that the policy can combine with vision and robot state. Before observing outcomes, we fixed a paired test that isolates this conversion. On canonical episodes 30–39, disjoint from the instruction-necessity episodes, both arms received the identical correct 32-token ID tensor. The ablated arm replaced only the output of the token-embedding lookup with exact zeros. Sequence length, token positions, vision, robot state, embodiment, the separate learned common BOS, action queries, transformer weights, and action head were preserved and hash-checked.

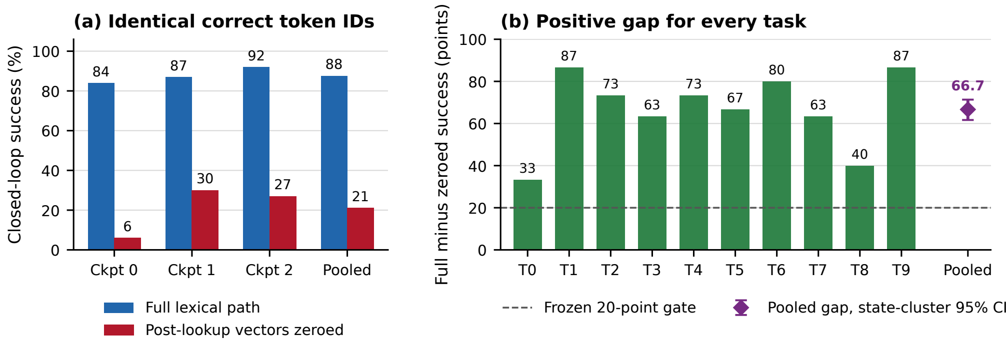

Figure 6: **Learned lexical-embedding-path necessity on familiar tasks.** Left: closed-loop success for the intact path and exact post-lookup zeroing, using 100 paired states per checkpoint and 300 pooled pairs. Right: every task aggregate has a positive paired gap. The pooled marker shows the frozen task-stratified state-cluster 95% bootstrap interval.

The intact path succeeds on $263/300 = 87.7\%$ of checkpoint-state pairs, compared with $63/300 = 21.0\%$ after zeroing the learned token vectors. The paired gap is 66.7 points, with 203 intact-only and 3 zeroed-only outcomes. Full-path success is 84%, 87%, and 92% by checkpoint, and the corresponding gaps are 78, 57, and 65 points. All ten task aggregates are positive. A pooled one-sided exact paired test gives $p = 1.42 \times 10^{-56}$, but this test is not cluster-robust because checkpoints reuse the same 100 states. The frozen 20,000-draw task-stratified state-cluster bootstrap interval is $[61.7, 71.3]$ points and is our cluster-aware inference. All eight prospectively fixed gates pass.

This establishes that the learned lexical-embedding path is necessary for these familiar LIBERO-Object tasks and three fixed policies. It does not establish semantic grounding, paraphrase robustness, unseen-object or compositional generalization, cross-suite or physical transfer, or a benefit unique to tensor decomposability.

A.2 Reproducibility details

Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. The anonymous artifact releases raw structural-certificate JSONs and all 2,000 episode rows behind the ViT member and ensemble results. Immutable SHA-256 manifests and standard-library verifiers independently recompute the capability totals, matched conventional comparison, four-suite ensemble and retention arithmetic, visual-bottleneck counts and intervals, and the attention, FFN, complete-block, modality-partition, full learned-forward, and RationalNorm certificate decisions. Preflight checks fail if canonical initial states cannot be loaded.

A.3 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.4 Prospectively fixed-rank visual bottleneck

The behavioral intervention acts at the width-384 visual-to-joint interface and keeps rank 96. Projector construction uses official Object tasks 0–3. Rank 96 was fixed before any corrected-basis outcome was observed, and the frozen evaluation uses tasks 8–9. Across three checkpoints and 300 canonical episodes, the full policy succeeds on 272 trials. The action-Jacobian projector retains $259/272 = 95.2\%$ of those successes, with a paired checkpoint-task-episode resampling interval of 91.2–98.6%. The activation-energy projector retains $267/272 = 98.2\%$, with an interval of 95.8–100.0%. Three fixed equal-rank random projectors retain $37/272 = 13.6\%$, $9/272 = 3.3\%$, and $22/272 = 8.1\%$.

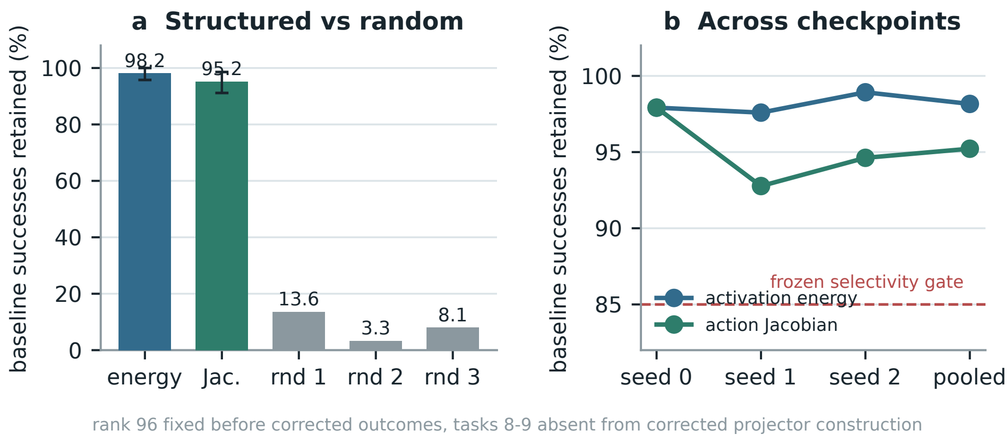

Figure 7: **Corrected closed-loop visual bottleneck.** Both learned rank-96 projectors preserve nearly all baseline successes across three checkpoints, while equal-rank random projectors do not. Error bars in panel (a) are deterministic paired cluster-resampling intervals. Tasks 8–9 are absent from corrected projector construction, but not from policy training. This is not a clean rank-selection-holdout result.

The action-Jacobian basis does not outperform the matched activation-energy basis. The stronger action-specificity gate therefore does not pass. The supported claim is structured visual sufficiency relative to random projectors. This intervention is data-derived and remains separate from the exact weight-only attention decomposition.

A.5 Numerical scope of rational conversion

The deployed degree-$[2/2]$ least-squares rational fit is exact as a rational computation even though it approximates RMS normalization. Its numerator has coefficients $(5.32997, 7.53552, 0.327274)$ and the denominator has coefficients $(1, 9.64827, 2.60664)$. Every denominator coefficient is positive, which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances, each invoked once per forward pass, on $4{,}096$ deterministic inputs for each of three convolutional checkpoints. Across $544{,}542{,}720$ activation rows, the minimum observed denominator is 1.0016 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3} gate. This primary certificate concerns pole-free denominators and floating-point fidelity to the deployed rational operator. It does not claim uniform agreement with exact RMS scaling or measure matched alternative-normalizer runtime overhead.

The reconstruction ladder extends from individual attention heads through FFNs and per-block replay to a separate sequential learned-forward certificate. That final record starts at exact prepared model tensors and reaches the linear action head on 48 fixed inputs. It is numerical and input-conditioned, not a compact symbolic contraction or an input-general full-policy proof. The ViT source-partition audit remains layerwise and descriptive. Figure 8 reports the frozen numerical gates and the depth profile without assigning causal importance.

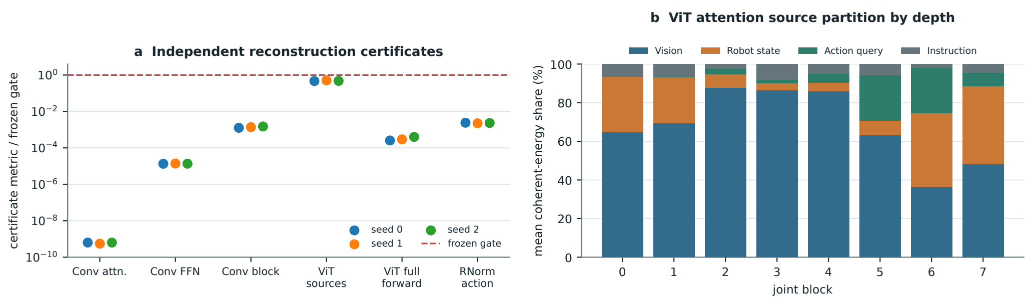

Figure 8: **Frozen reconstruction ladder and descriptive ViT source partition.** Panel (a) reports observed-to-gate ratios for independent Conv attention, joint-FFN, and complete joint-block reconstruction, the ViT source-partition identity, the sequential ViT learned-forward action replay, and primary RationalNorm action fidelity. All plotted primary numerical gates pass. Panel (b) shows descriptive mean coherent-energy shares after the attention output projection across 384 checkpoint-input cases per joint block. The depth trend is not a causal importance or grounding result. The learned-forward replay excludes preprocessing, fixed action unnormalization, chunk execution, simulator dynamics, and success.